



Caringbah High School

Year 12 2017

Mathematics Extension 2

HSC Course

Assessment Task 4

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
- Board-approved calculators may be used
- The reference sheet from the Board of Studies is provided.
- In Questions 11-16, show relevant mathematical reasoning and/or calculations
- Marks may not be awarded for partial or incomplete answers

Total marks – 100

Section I 10 marks

Attempt Questions 1-10
Mark your answers on the answer sheet provided. You may detach the sheet and write your name on it.

Section II 90 marks

Attempt Questions 11-16
Write your answers in the answer booklets provided. Ensure your name or student number is clearly visible.

Name: _____

Class: _____

Marker's Use Only							
Section I	Section II						Total
Q 1-10	Q11	Q12	Q13	Q14	Q15	Q16	
/10	/15	/15	/15	/15	/15	/15	/100

Section I

10 marks

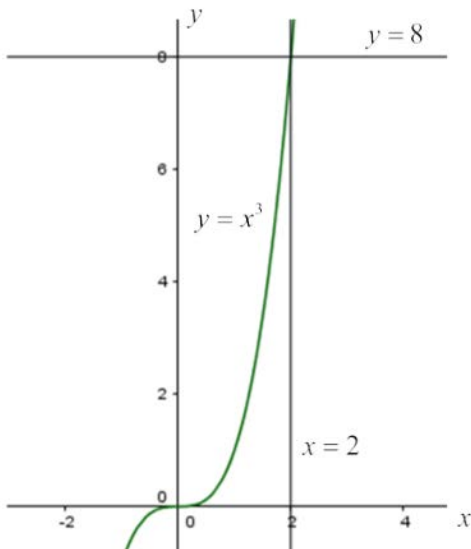
Attempt Questions 1-10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10

1. For the hyperbola $\frac{x^2}{9} - \frac{y^2}{72} = 1$ the eccentricity is
- (A) $\frac{3}{2\sqrt{2}}$
- (B) 3
- (C) $2\sqrt{2}$
- (D) 8
2. If ω is one of the complex roots of $z^3 - 1 = 0$, then the value of $\frac{1}{1+\omega} + \frac{1}{1+\omega^2}$ is
- (A) -1
- (B) 2
- (C) 0
- (D) 1

3. The volume of the solid generated when the region bounded by $y = x^3$, $x = 0$, $y = 8$ is rotated about the line $x = 2$ is given by



- (A) $\pi \int_0^8 4x - x^2 dx$
- (B) $\pi \int_0^8 4y^{\frac{1}{3}} - y^{\frac{2}{3}} dy$
- (C) $\pi \int_0^2 4x - x^2 dx$
- (D) $\pi \int_0^2 4y^{\frac{1}{3}} - y^{\frac{2}{3}} dy$
4. The polynomial equation $x^3 - 2x^2 + 3 = 0$ has roots α , β and γ . What is the value of $\alpha^3 + \beta^3 + \gamma^3$?
- (A) -2
- (B) -1
- (C) -8
- (D) 8

5. If $x = \theta - \sin \theta$ and $y = 1 - \cos \theta$ which of the following is an expression for $\frac{dy}{dx}$?

(A) $\cot^2 \frac{\theta}{2}$

(B) $\cot \frac{\theta}{2}$

(C) $\tan \frac{\theta}{2}$

(D) $\tan^2 \frac{\theta}{2}$

6. Let α, β and γ be roots of the equation $x^3 + 3x^2 + 4 = 0$. Which of the following polynomial equations have roots α^2, β^2 and γ^2 ?

(A) $x^3 - 9x^2 - 24x - 4 = 0$

(B) $x^3 - 9x^2 - 12x - 4 = 0$

(C) $x^3 - 9x^2 - 24x - 16 = 0$

(D) $x^3 - 9x^2 - 12x - 16 = 0$

7. Which of the following is an expression for $\int x e^{\frac{x}{2}} dx$?

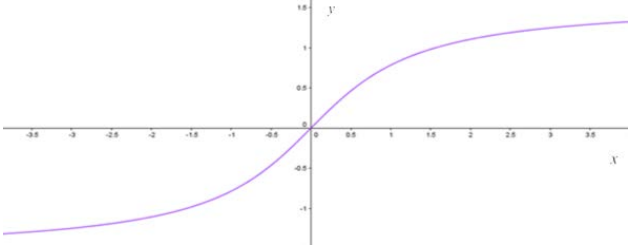
(A) $\frac{1}{2} x e^{\frac{x}{2}} - \frac{1}{4} e^{\frac{x}{2}} + c$

(B) $\frac{1}{2} x e^{\frac{x}{2}} - \frac{1}{2} e^{\frac{x}{2}} + c$

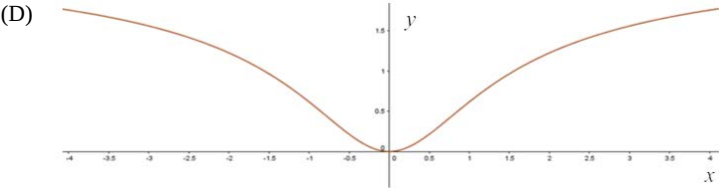
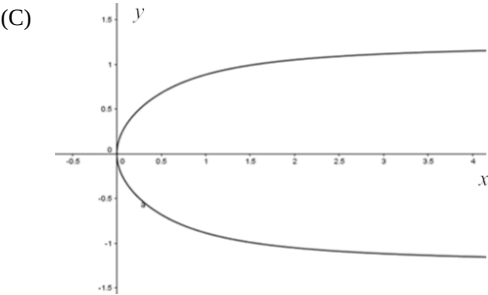
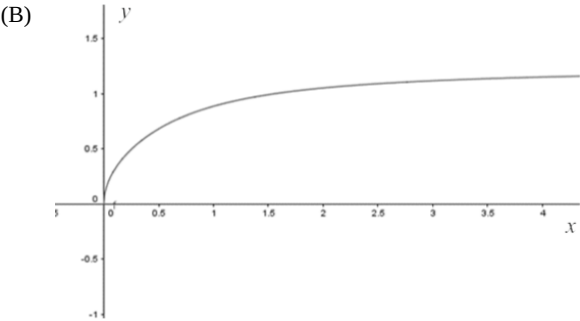
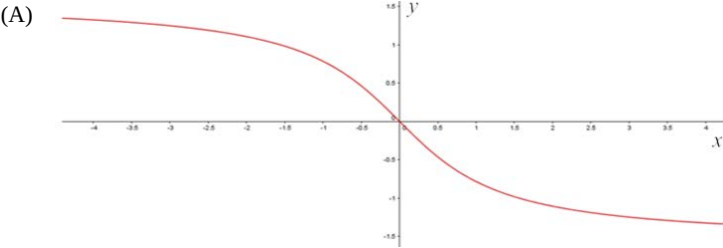
(C) $2 x e^{\frac{x}{2}} - 2 e^{\frac{x}{2}} + c$

(D) $2 x e^{\frac{x}{2}} - 4 e^{\frac{x}{2}} + c$

8. The diagram shows the graph of the function $y=f(x)$



Which of the following is the graph of $y^2 = f(x)$



9. It is given that $3+i$ is a root of $P(z) = z^3 + az^2 + bz + 10$, where a and b are real numbers. Which expression factorises $P(z)$ over the set of real numbers?
- (A) $(z+1)(z^2 - 6z + 10)$
(B) $(z-1)(z^2 - 6z - 10)$
(C) $(z+1)(z^2 + 6z + 10)$
(D) $(z+1)(z^2 + 6z - 10)$
10. z and w are two complex numbers. Which of the following statements is always true?
- (A) $|z| - |w| \geq |z + w|$
(B) $|z| + |w| \leq |z + w|$
(C) $|z| + |w| \leq |z + w|$
(D) $|z + w| + |z| \geq |w|$

End of Section I

Section II

60 marks

Attempt Questions 11–16

Allow about 2 hours 45 minutes for this section

Answer each question in a SEPARATE writing booklet.

Your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use a SEPARATE writing booklet.

(a) Use the substitution $u = 1 + \sin^2 x$ to find $\int \frac{\sin 2x}{\sqrt{1 + \sin^2 x}} dx$ 2

(b) Evaluate $\int_{\sqrt{2}}^{\sqrt{3}} \frac{2e^{\frac{2}{x^2}}}{x^3} dx$ 3

(c) (i) Find real numbers a , b and c such that 2

$$\frac{9-x}{(1+x^2)(1+x)} = \frac{ax+b}{1+x^2} + \frac{c}{1+x}$$

(ii) Hence, or otherwise, find 3

$$\int \frac{9-x}{(1+x^2)(1+x)} dx$$

(d) Evaluate $\int_0^{\frac{\pi}{2}} e^x \sin 2x dx$ 5

End of Question 11

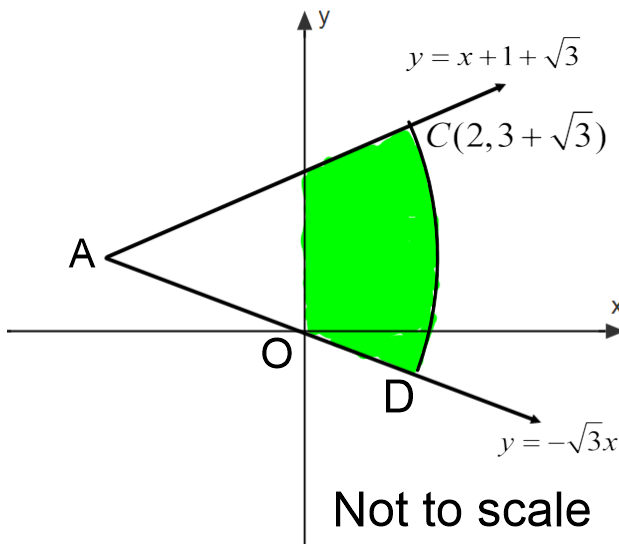
Question 12 (15 marks) Use a SEPARATE writing booklet.

- (a) (i) Express $z = -1 + \sqrt{3}i$ in mod-arg form. 1
- (ii) Hence find $(-1 + \sqrt{3}i)^5$, giving your answer in the form $a + ib$ 2

- (b) (i) Find the square roots of the complex number i . 2
Express your answers in the form $z = a + ib$,
where a and b are real numbers.

- (ii) Given that $z = \frac{2(\sqrt{2} - 2\sqrt{i})}{1 - i}$ and $\text{Im}(z) < 0$, using the solution to (i), 3
express z in the form $a + ib$, where a and b are real numbers.

- (c) 4



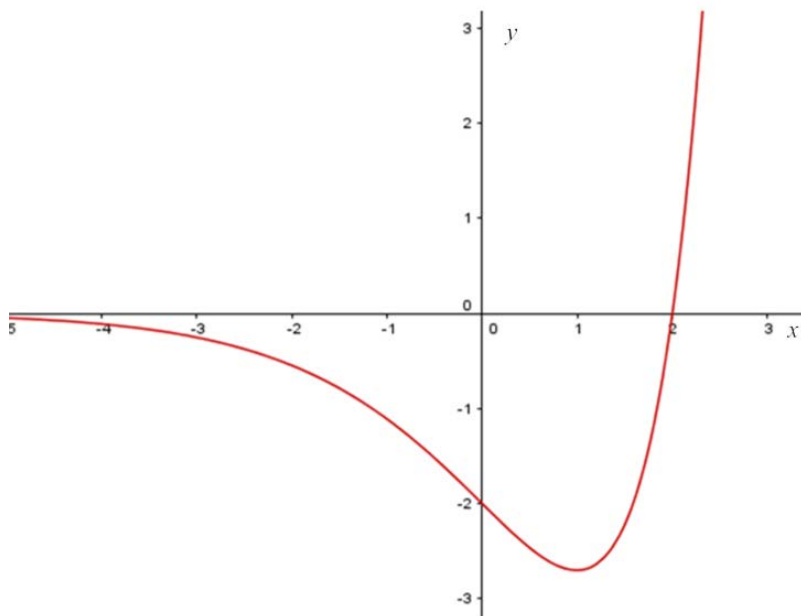
In the Argand diagram above, the shaded region is part of a circle centred at A, with radii AD and AC. Find the conditions that should be satisfied by the complex numbers z which are in this region.

- (d) If x , y and z are positive and unequal real numbers, prove that
- (i) $x + y - 2\sqrt{xy} > 0$ 1
- (ii) $(x + y)(y + z)(z + x) > 8xyz$ 2

End of Question 12

Question 13 (5 marks) Use a SEPARATE writing booklet.

- (a) The diagram show the graph of the function $y = f(x)$ where $f(x) = e^x(x-2)$



On separate $\frac{1}{2}$ page diagrams sketch the following graphs showing the coordinates of any turning points and the equations of any asymptotes and the x and y intercepts where possible.

- | | | |
|-------|---------------------------------|---|
| (i) | $y = f(x)$ | 1 |
| (ii) | $y = \ln[f(x)]$ | 2 |
| (iii) | $y = e^{f(x)}$ | 2 |
| (iv) | $y = f\left(\frac{1}{x}\right)$ | 2 |

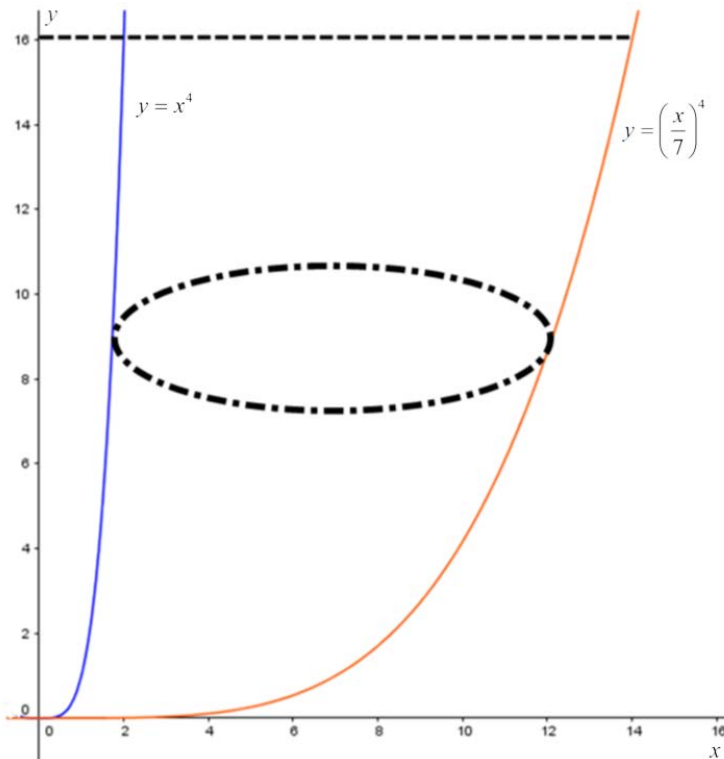
- (b) Consider the curve defined by the equation $3x^2 + y^2 - 2xy - 8x + 2 = 0$

- | | | |
|------|---|---|
| (i) | Show that $\frac{dy}{dx} = \frac{3x - y - 4}{x - y}$ | 2 |
| (ii) | Find the coordinates of the points on the curve where the tangent to the curve is parallel to the line $y = 2x$ | 2 |

Question 13 continues on page 9

(c)

A mould for a drinking horn is bounded by the curves $y = x^4$ and $y = \left(\frac{x}{7}\right)^4$ between $y = 0$ and $y = 16$. Each cross section perpendicular to the y axis is a circle. All measurements are in cm. Find the exact volume of the drinking horn in terms of $\pi \text{ cm}^3$



End of Question 13

Question 14 (15 marks) Use a SEPARATE writing booklet.

- (a) (i) Using the result of de Moivre's theorem that $\cos 3\theta = 4\cos^3 \theta - 3\cos \theta$, 2
solve the equation $8x^3 - 6x + 1 = 0$

(ii) Hence deduce that $\alpha) \cos \frac{2\pi}{9} + \cos \frac{4\pi}{9} = \cos \frac{\pi}{9}$ 1

and $\beta) \sec \frac{\pi}{9} \sec \frac{2\pi}{9} \sec \frac{4\pi}{9} = 8$ 1

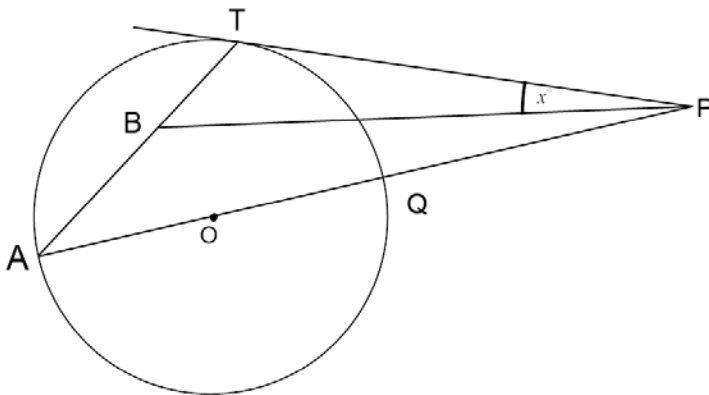
- (b) Consider the equation $P(z) = z^4 + pz^3 + qz + r$ when $P(z) = 0$, p , q and r are real numbers. The sum of the roots of this equation is 6 more than the product of the roots. If $1+i$ is a root of the equation,

(i) form 2 equations in terms of p , q that satisfy $P(z) = 0$ 3

(ii) Hence find the values of p , q and r . 2

(iii) Find all the roots of the equation 2

- (c) PT is a tangent to a circle centre O. PO produced meets the circle at A and Q. 4
The bisector of $\angle TPA$ meets the chord AT at B. Let $\angle TPB = x^\circ$

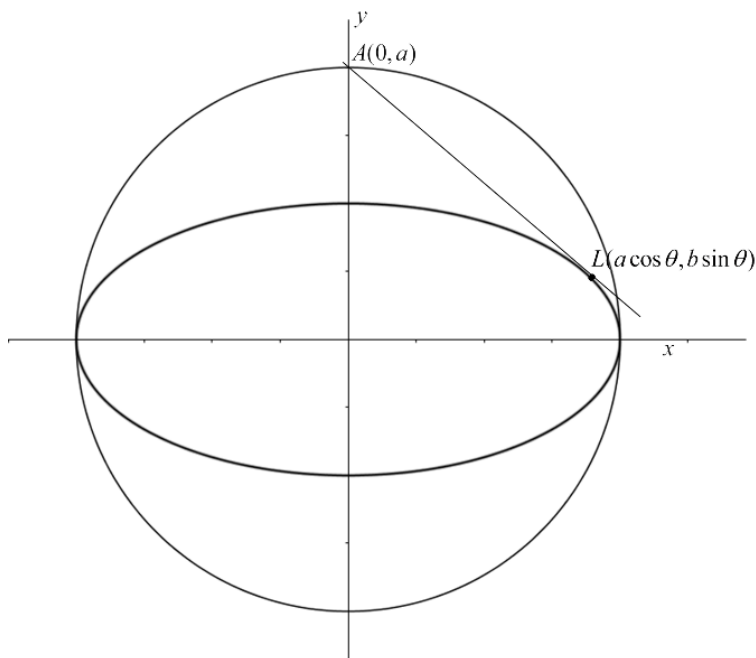


Use the template provided to prove that $\angle PBT = 45^\circ$.

End of Question 14

Question 15 (15 marks) Use a SEPARATE writing booklet.

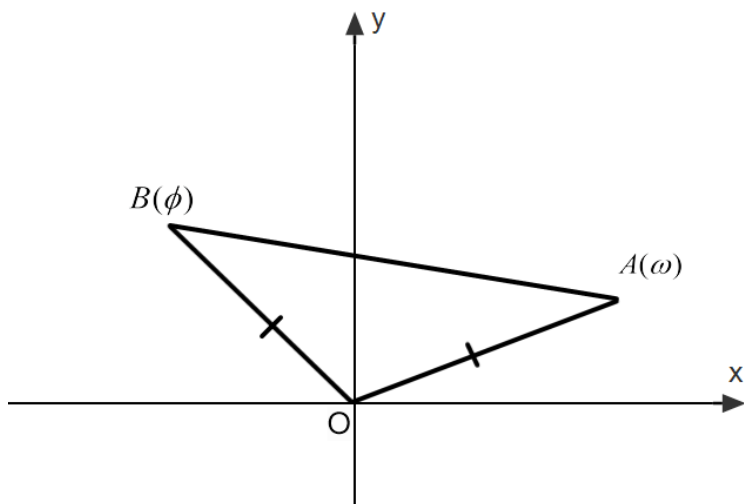
- (a) $L(a \cos \theta, b \sin \theta)$ is a variable point on the ellipse with equation $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.



- (i) Show that the equation of the tangent at L is $\frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1$ 2
- (ii) This tangent meets the auxiliary circle of the ellipse as shown on the minor axis at $A(0, a)$. Show that the line from L that is perpendicular to the x axis passes through a focus of the ellipse. 4
- (b) (i) Show that the equation of the normal to the rectangular hyperbola $xy = 4$ at the point $P(2p, \frac{2}{p})$ is $py - p^3x = 2(1 - p^4)$. 2
- (ii) If this normal meets the hyperbola again at the point $Q(2q, \frac{2}{q})$ show that $p^3q = -1$ 2

Question 15 continues on page 12

- (b) The Argand diagram shows the points A and B which represent the complex numbers ω and ϕ respectively. Given that $\triangle BOA$ is isosceles and $\angle BOA = \frac{2\pi}{3}$ 5



Show that
 $(\omega + \phi)^2 = \omega\phi$

End of Question 15

Question 16 (15 marks) Use a SEPARATE writing booklet.

- (a) (i) On the same set of axes sketch the graphs of 2
 $y = \frac{1}{x+1}$ and $y = \frac{1}{x+2}$

- (ii) The region bounded by the curves $y = \frac{1}{x+1}$, $y = \frac{1}{x+2}$, the ordinates 5
 $x = 0$ and $x = 2$ is rotated about the y axis. Using the method of
cylindrical shells, calculate the volume of the solid so generated.

- (b) Let $I_n = \int_0^{\frac{1}{2}} \frac{1}{(1+4x^2)^n} dx$, where n is a positive integer.

- (i) Find the value of I_1 2

- (ii) Using integration by parts, show that 4

$$I_n = \frac{2nI_{n+1}}{2n-1} + \frac{1}{2^{n+1}(1-2n)}$$

$$\left(\text{Hint : } \frac{m}{(1+m)^{n+1}} = \frac{1+m-1}{(1+m)^{n+1}} \right)$$

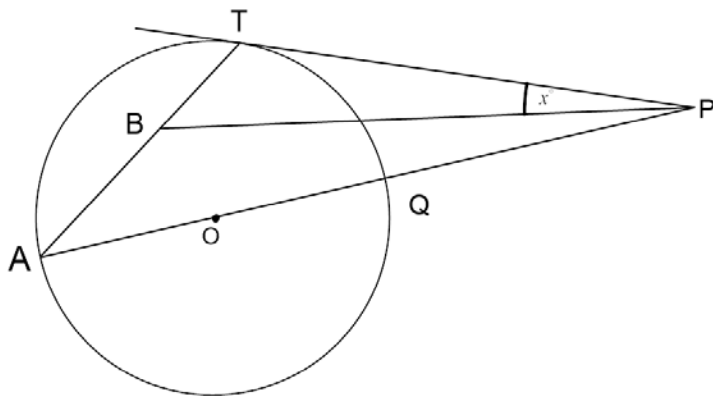
- (iii) Hence evaluate $I_3 = \int_0^{\frac{1}{2}} \frac{1}{(1+4x^2)^3} dx$ 2

End of Paper

Q14 (c) Answer Template

Name: _____

PT is a tangent to a circle centre O. PO produced meets the circle at A and Q. The bisector of $\angle TPA$ meets the chord AT at B. Let $\angle TPB = x^\circ$



Prove that $\angle PBT = 45^\circ$.



Caringbah High School

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Student Number

Examination: Ex 2 2017 TRIAL

MULT. Choice: 1 B 2 D 3 B 4 B 5 B 6 C 7 D 8 C 9 A 10 D

1. $e^2 = \frac{b^2}{a^2} + 1 = \frac{72}{9} + 1 = 9$ $e = 3$ [B]

2. $\frac{1}{w+1} + \frac{1}{1+w^2} = \frac{1+w^2+1+w}{(1+w)(1+w^2)}$ $1+w+w^2=0$
 $= \frac{1}{1+w+w^2+w^3} = \frac{1}{1} = D$

3. $\pi \int_0^8 2 - (2-x)^2 dy = \pi \int_0^8 4 - 4 + 4x - x^2 dx = \pi \int_0^8 4x - x^2 dx$ [B]
 $= \pi \int_0^8 4y^{\frac{1}{2}} - y^{\frac{2}{3}} dy$

4. $x^3 - 2x^2 + 3 = 0$

$2x + \beta + \gamma = 2$ $2\beta + 2\gamma + \alpha\gamma = 0$ $2\beta\gamma = -3$

$\alpha^3 = 2\alpha^2 - 3$ so $\alpha^3 + \beta^3 + \gamma^3 = 2(\alpha^2 + \beta^2 + \gamma^2) - 9$

$2\beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)$
 $= 4$

$\therefore \alpha^3 + \beta^3 + \gamma^3 = 2 \times 4 - 9 = -1$ [B]

5. $x = \theta - \sin \theta$ $y = 1 - \cos \theta$

$\frac{dx}{d\theta} = 1 - \cos \theta$ $\frac{dy}{d\theta} = \sin \theta$

$\frac{dy}{dx} = \frac{dy}{d\theta} \times \frac{d\theta}{dx} = \frac{\sin \theta}{1 - \cos \theta} = \frac{2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}}{1 - (1 - 2 \sin^2 \frac{\theta}{2})} = \frac{\cos \frac{\theta}{2}}{\sin \frac{\theta}{2}} = \cot \frac{\theta}{2}$ [B]

6. $x = \alpha^2$ so $\alpha = \sqrt{x}$

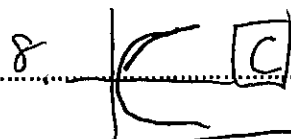
$x^{\frac{3}{2}} + 3x + 4 = 0$ [C]

$3x + 4 = -x^{\frac{3}{2}}$

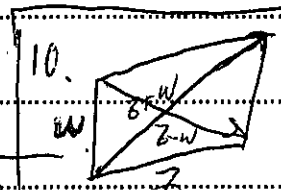
$9x^2 + 24x + 16 = x^3$

$\therefore x^3 - 9x^2 - 24x - 16 = 0$

7. $\int x e^{\frac{x}{2}} dx = 2x e^{\frac{x}{2}} - \int 2e^{\frac{x}{2}} dx$ [D]
 $= 2x e^{\frac{x}{2}} - 4e^{\frac{x}{2}} + C$



9. $(3+\gamma)(3-\gamma) \times 8 = -10$
 $10\gamma = -10$
 $\gamma = -1$
 $\therefore (3+1)(3^2 - 6\gamma + 10)$ [A]



10. $|3+w| + |3| \geq |w|$ [D]



Examination:.....

1/a) $u = 1 + \sin^2 x$

$du = 2 \sin x \cos x dx$

$$\int \frac{\sin 2x}{\sqrt{1 + \sin^2 x}} dx$$

$$= \sin 2x$$

$$\int \frac{du}{\sqrt{u}} = 2u^{\frac{1}{2}} + C = 2\sqrt{1 + \sin^2 x} + C$$

b) $\int_{\sqrt{2}}^{\sqrt{3}} \frac{2e^{\frac{2}{x}}}{x^3} dx = \int_{\sqrt{2}}^{\sqrt{3}} 2(e^{2x^{-2}}) x^{-3} dx$

$$= \left[-\frac{e^{2x^{-2}}}{2} \right]_{\sqrt{2}}^{\sqrt{3}} = -\frac{e^{\frac{2}{3}}}{2} + \frac{e^1}{2} = \frac{e - e^{\frac{2}{3}}}{2}$$

c) i) $\frac{9-x}{(1+x)(1+x^2)} = \frac{ax+b}{1+x} + \frac{c}{1+x^2}$

$$(a+b)(1+x) + c(1+x^2) = 9-x$$

$$\text{Let } x = -1 \quad c = 5 \quad \text{Let } x = 0 \quad b = 4 \quad \text{Let } x = 1 \quad a = -5$$

$$\frac{9-x}{(1+x)(1+x^2)} = \frac{-5x+4}{x^2+1} + \frac{5}{x^2+1}$$

ii) $\int \frac{9-x}{(1+x)(1+x^2)} dx = \int \frac{4}{x^2+1} dx - \int \frac{5x}{x^2+1} dx + \int \frac{5}{x^2+1}$

$$= 4 \tan^{-1} x - \frac{5}{2} \ln(x^2+1) + 5 \ln|x+1| + C$$

$$= 4 \tan^{-1} x - \frac{5}{2} \ln \left(\frac{x^2+1}{(x+1)^2} \right) + C$$

$$\text{or } = 4 \tan^{-1} x + \frac{5}{2} \ln \left(\frac{(x+1)^2}{x^2+1} \right) + C$$

d) $\int_0^{\frac{\pi}{2}} e^x \sin 2x dx = \left[-\frac{1}{2} e^x \cos 2x \right]_0^{\frac{\pi}{2}} + \frac{1}{2} \int_0^{\frac{\pi}{2}} e^x \cos 2x dx$

$$= \frac{e^{\frac{\pi}{2}}}{2} + \frac{1}{2} + \left[\frac{1}{4} e^x \sin 2x \right]_0^{\frac{\pi}{2}} - \frac{1}{4} \int_0^{\frac{\pi}{2}} e^x \sin 2x dx$$

$$\therefore \frac{5}{4} \int_0^{\frac{\pi}{2}} e^x \sin 2x dx = \frac{e^{\frac{\pi}{2}}}{2} + 1$$

$$\therefore \int_0^{\frac{\pi}{2}} e^x \sin 2x dx = \frac{2}{5} (e^{\frac{\pi}{2}} + 1)$$



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Student Number

Examination:.....

Q12 a) (i) $z = -1 + \sqrt{3}i$ ~~$2 \text{cis } \frac{2\pi}{3}$~~ $2 \text{cis } \frac{2\pi}{3}$
 $r = \sqrt{1+3} = \sqrt{4} = 2$

ii) $(-1 + \sqrt{3}i)^5 = 32 \text{cis } \frac{10\pi}{3} = 32 \text{cis } \frac{4\pi}{3} = 32 \times \left(-\frac{1}{2} - \frac{\sqrt{3}i}{2}\right) = -16 - 16\sqrt{3}i$

b) $\sqrt{i} = \text{cis}\left(\frac{\pi}{2} + 2k\pi\right) \quad k=0,1$ or $(x+iy)^2 = i$

$= \text{cis } \frac{\pi}{4}, \text{cis } \frac{5\pi}{4}$

$= \frac{1+i}{\sqrt{2}}, \frac{-1-i}{\sqrt{2}}$

$x^2 - y^2 = 0$

$2xy = 1$

$x^2 - \frac{1}{4x^2} = 0$

$y = \frac{1}{2x}$

$4x^4 - 1 = 0$

$x = \pm\sqrt{2}, y = \pm\frac{1}{\sqrt{2}}$

$(2x^2-1)(2x^2+1) = 0$

$z = \pm \frac{1+i}{\sqrt{2}}$

ii) $z = \frac{2(\sqrt{2} - 2(\sqrt{i}))}{1-i}$

$= \frac{2(\sqrt{2} - 2(\frac{1+i}{\sqrt{2}}))}{(1-i)}$

$= \frac{2(\sqrt{2} \pm (\sqrt{2} + \sqrt{2}i))}{(1-i)}$

$= \frac{2(-\sqrt{2}i)(1+i)}{(1-i)(1+i)} \text{ or } \frac{2(2\sqrt{2} + \sqrt{2}i)(1+i)}{(1-i)(1+i)}$

$= \frac{2(\sqrt{2} - \sqrt{2}i)}{2} , \frac{2(\sqrt{2} + 3\sqrt{2}i)}{2}$

$= \sqrt{2} - \sqrt{2}i , \sqrt{2} + 3\sqrt{2}i$

but $\text{Im}(z) < 0$ so only solution

is $\sqrt{2} - \sqrt{2}i$



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Student Number

Examination:.....

$$c) \quad x+1+\sqrt{3} = -\sqrt{3}x$$

$$x + \sqrt{3}x = -(1+\sqrt{3})$$

$$x(1+\sqrt{3}) = -(1+\sqrt{3})$$

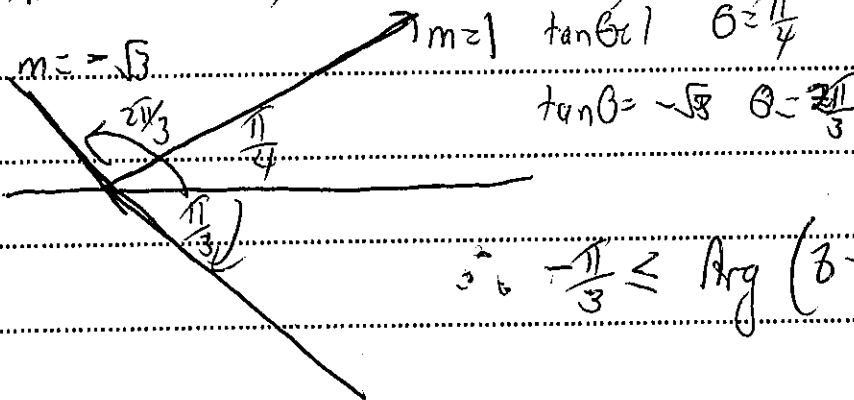
$$\therefore x = -1, \quad y = -\sqrt{3}x - 1 = \sqrt{3}$$

$$A(-1, \sqrt{3}) \quad C(2, 3+\sqrt{3})$$

circle centre $(-1, \sqrt{3})$ radius $AC: AC = \sqrt{3^2 + 3^2} = 3\sqrt{2}$.

$$\therefore |z - (-1 + \sqrt{3}i)| \leq 3\sqrt{2}$$

$$\text{and } \operatorname{Re}(z) \geq 0 \quad (\text{or } x \geq 0)$$



$$\therefore -\frac{\pi}{3} \leq \operatorname{Arg}(z - (-1 + \sqrt{3}i)) \leq \frac{\pi}{4}$$

$$i) \quad x + y - 2\sqrt{xy} > 0$$

$$x, y > 0 \text{ and } x \neq y$$

$$\text{So } (\sqrt{x} - \sqrt{y})^2 > 0$$

$$\therefore x - 2\sqrt{xy} + y > 0$$

$$\therefore x + y - 2\sqrt{xy} > 0$$

ii) similarly

$$y + z \geq 2\sqrt{yz}$$

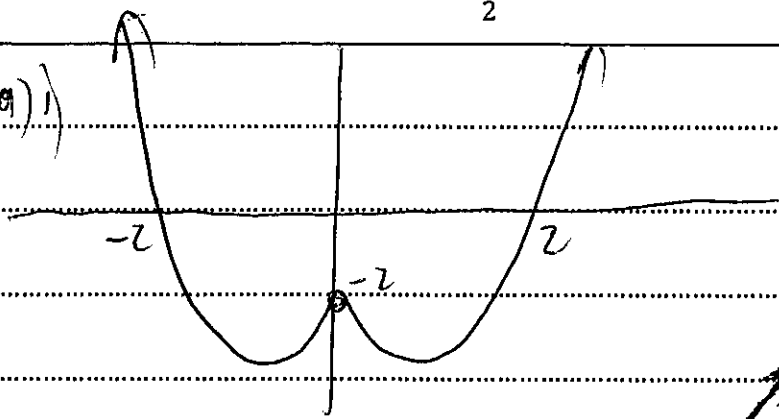
$$z + x \geq 2\sqrt{zx}$$

$$\therefore (x+y)(y+z)(z+x) \geq 2\sqrt{xy} \times 2\sqrt{yz} \times 2\sqrt{zx}$$

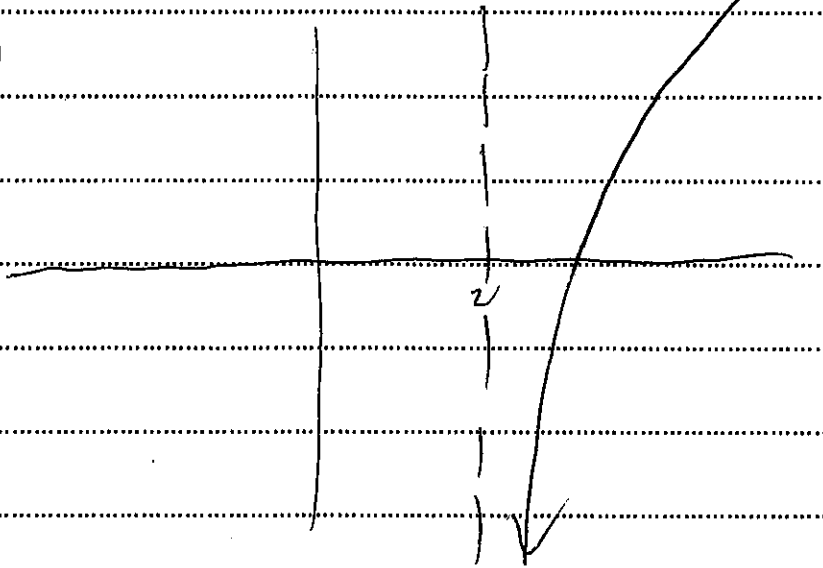
$$\geq 8\sqrt{x^2y^2z^2}$$

$$\geq 8xyz$$

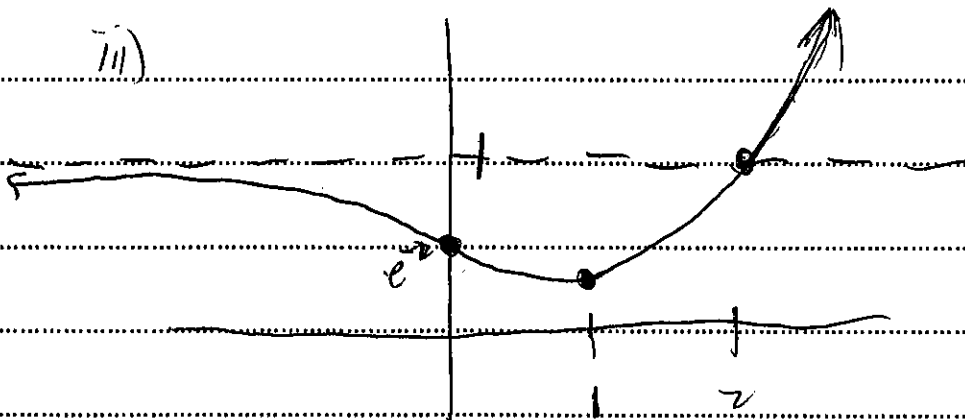
13) a) i)



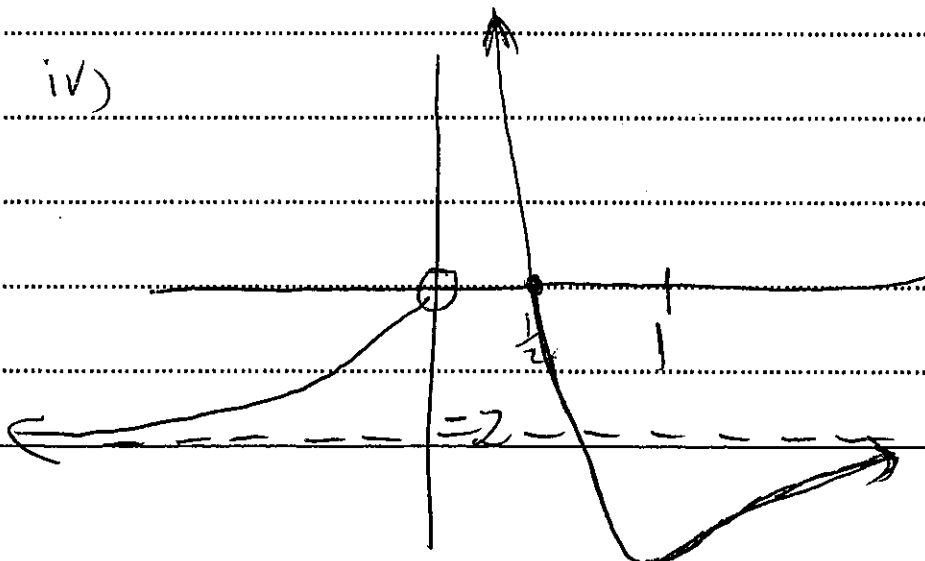
ii)



iii)



iv)



$$b) \quad 3x^2 + y^2 - 2xy - 8x + 2 = 0$$

$$6x + 2y \frac{dy}{dx} - 2y - 2x \frac{dy}{dx} - 8 = 0$$

$$6x - 2y - 8 + (2y - 2x) \frac{dy}{dx} = 0$$

$$(2y - 2x) \frac{dy}{dx} = 3x - y - 4$$

$$\therefore \frac{dy}{dx} = \frac{3x - y - 4}{x - y}$$

~~$$\frac{dy}{dx} = \frac{3x - y - 4}{x - y}$$~~

~~$$3x - y - 4 = 2x - 2y$$~~

~~$$3x = 2x$$~~

~~$$x = \frac{4}{3}$$~~

~~$$y = \frac{8}{3}$$~~

$$\frac{3x - y - 4}{x - y} = 2$$

$$3x - y - 4 = 2x - 2y$$

$$y = 4 - x$$

$$\therefore 3x^2 + (4 - x)^2 + 2x(4 - x) - 8x + 2 = 0$$

$$6x^2 - 24x + 18 = 0$$

$$6(x - 3)(x - 1) = 0$$

$$\therefore \text{At } P: \quad y = 4 - x \quad x = 3 \text{ or } x = 1$$

So
tangent at $(3, 1)$ $(1, 3)$ are $\parallel$ to $y = 2x$

$$c) \text{ - Diameter } = x_2 - x_1 = 7y^{\frac{1}{4}} - y^{\frac{1}{4}} = 6y^{\frac{1}{4}} \text{ So } r = 3y^{\frac{1}{4}}$$

$$\text{Area} = \pi (3y^{\frac{1}{4}})^2 = 9\pi y^{\frac{1}{2}}$$

$$\text{So } \delta V = 9\pi y^{\frac{1}{2}} \delta y$$

$$\therefore V = \lim_{\delta y \rightarrow 0} \sum_{y=0}^{16} 9\pi y^{\frac{1}{2}} \delta y \therefore V = 9\pi \int_0^{16} y^{\frac{1}{2}} dy$$
$$= 6\pi y^{\frac{3}{2}} \Big|_0^{16} = 384\pi \text{ cm}^3$$



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Student Number

Examination:.....

Q14 i) $x = \cos \theta$

$$8 \cos^3 \theta - 6 \cos \theta + 1 = 0$$

$$4 \cos^3 \theta - 3 \cos \theta = -\frac{1}{2}$$

$$\cos 3\theta = -\frac{1}{2}$$

$$3\theta = \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{8\pi}{3}, \frac{10\pi}{3}, \frac{14\pi}{3}, \frac{16\pi}{3}$$

$$\theta = \frac{2\pi}{9}, \frac{4\pi}{9}, \frac{8\pi}{9}, \frac{10\pi}{9}, \frac{14\pi}{9}, \frac{16\pi}{9}$$

$$\text{So } x = \cos \theta = \cos \frac{2\pi}{9}, \cos \frac{4\pi}{9}, \cos \frac{8\pi}{9}, \cos \frac{10\pi}{9}, \cos \frac{14\pi}{9}, \cos \frac{16\pi}{9}$$

$$= -\cos \frac{\pi}{9}, -\cos \frac{\pi}{9}, \cos \frac{4\pi}{9}, \cos \frac{4\pi}{9}, \cos \frac{2\pi}{9}, \cos \frac{2\pi}{9}$$

$$\therefore \text{Roots are } \cos \frac{2\pi}{9}, \cos \frac{4\pi}{9}, -\cos \frac{\pi}{9}$$

ii) $2 + \beta + \gamma = 0 \therefore \cos \frac{2\pi}{9} + \cos \frac{4\pi}{9} - \cos \frac{\pi}{9} = 0$

$$\therefore \cos \frac{2\pi}{9} + \cos \frac{4\pi}{9} = \cos \frac{\pi}{9}$$

iii) $2\beta\gamma = -\frac{1}{8} \quad \cos \frac{2\pi}{9} \times \cos \frac{4\pi}{9} \times -\cos \frac{\pi}{9} = -\frac{1}{8}$

$$\therefore \sec \frac{2\pi}{9} \times \sec \frac{4\pi}{9} \times \sec \frac{\pi}{9} = 8$$

Start here for
Question Number:

11

$$2 + 13 + 8 = -p$$

$$2 + 8 = r$$

$$(Q14b)i) -p = r + 6$$

$$(\sqrt{2} \cos \frac{\pi}{4})^4$$

$$(1+i)^4 = -4$$

$$(1-i)^4 = (\sqrt{2} \cos \frac{\pi}{4})^4$$

$$(1+i)^3 = (\sqrt{2} \cos \frac{\pi}{4})^3$$

$$= -4$$

$$= -2 + 2i$$

$$(1-i)^3 = (\sqrt{2} \cos \frac{\pi}{4})^3$$

$$= -2 - 2i$$

Using $z = 1+i$

$$-4 + p(-2+2i) + q(1+i) - p - 6 = 0$$

$$-10 - 3p + q + 2ip + iq = 0 \quad \text{--- (1)}$$

Using $z = 1-i$

$$-4 + p(-2-2i) + q(1-i) - p - 6 = 0$$

$$-10 - 3p + q - 2ip - iq = 0 \quad \text{--- (2)}$$

$$(1) + (2)$$

$$-20 - 6p + 2q = 0$$

$$\therefore -3p + q = 10$$

$$(1) - (2)$$

$$4ip + 2iq = 0$$

$$\therefore 2p + q = 0$$

$$\text{ii)} \quad -3p + q = 10$$

$$2p + q = 0$$

$$-5p = 10$$

$$p = -2$$

$$q = 4$$

$$r = +2 - 6 = -4$$

/2

iii) Equation is

$$z^4 - 2z^3 + 4z - 4 = 0$$

$$(z - (1+i))(z - (1-i)) = z^2 - 2z + 2 \text{ is a factor}$$

$$\begin{array}{r} z^2 - 2z + 2 \overline{) z^4 - 2z^3 + 4z - 4} \\ \underline{z^4 - 2z^3 + 2z^2} \\ -2z^2 + 4z - 4 \end{array}$$

/2.

$\therefore$ Roots are $(1+i)$ $(1-i)$ $\sqrt{2}$, $-\sqrt{2}$

Additional writing space on back page.

c)

$$\angle ATC = 90^\circ \text{ (}\angle \text{ in semi-circle)}$$

$$\angle PTC = \angle PAC = y \text{ (angle in alt segment)}$$

In $\triangle APT$

$$2x + 2y + 90 = 180$$

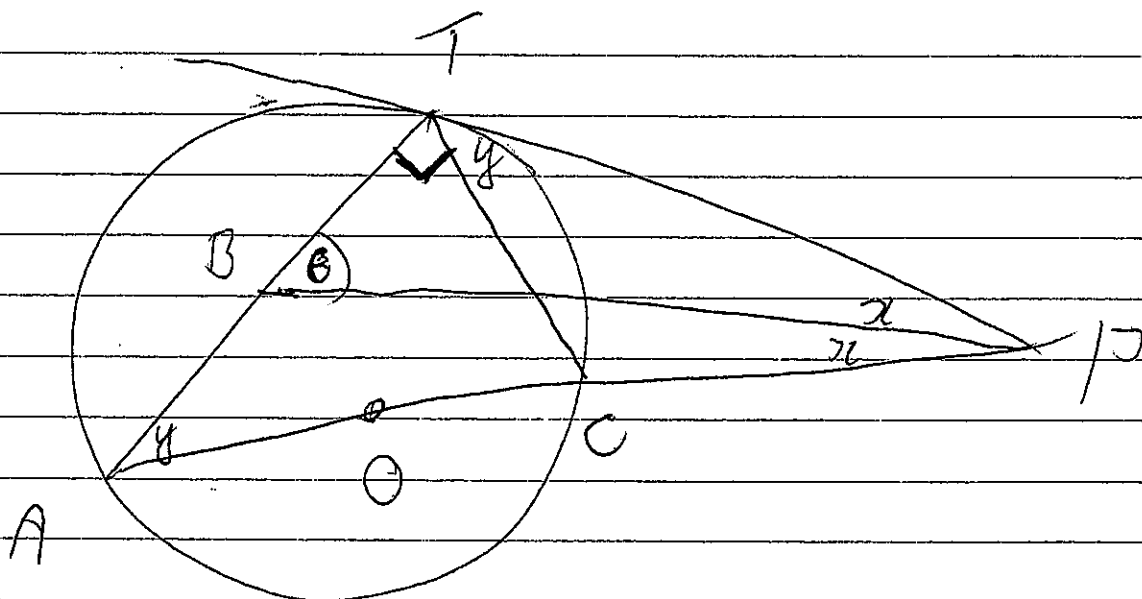
$$\therefore 2x + 2y = 90$$

$$\therefore x + y = 45$$

$\therefore$ In $\triangle TBP$

$$90 + x + y + \angle TBP = 180$$

$$\therefore \angle TBP = 180 - 135 = 45^\circ$$



Q15 a) i) $L(a \cos \theta, b \sin \theta)$

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \quad (a > b)$$

$$\frac{2x}{a^2} + \frac{2y}{b^2} \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = \frac{-b^2 x}{a^2 y}$$

$$= \frac{-b^2 a \cos \theta}{a^2 b \sin \theta}$$

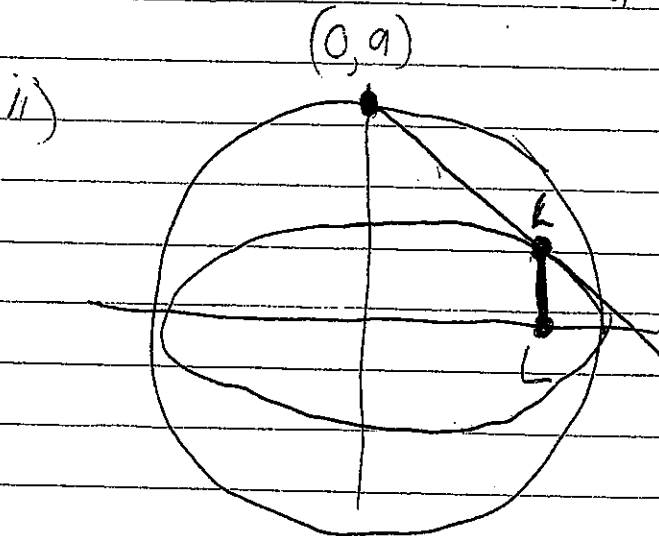
$$= \frac{-b \cos \theta}{a \sin \theta} \quad \downarrow$$

$$y - b \sin \theta = \frac{-b \cos \theta}{a \sin \theta} (x - a \cos \theta)$$

$$a y \sin \theta - a b \sin^2 \theta = -b x \cos \theta + a b \cos^2 \theta$$

$$b x \cos \theta + a y \sin \theta = a b (\sin^2 \theta + \cos^2 \theta) \quad \downarrow$$

$$= a b$$



$(0, \pm a)$ satisfies $b x \cos \theta + a y \sin \theta = a b$

$$\pm a^2 \sin \theta = a b$$

$$\sin \theta = \pm \frac{b}{a} \quad \downarrow$$

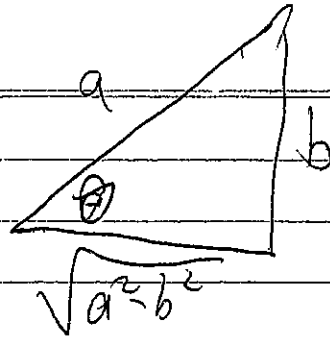
$$\therefore L(a \cos \theta, a \sin \theta)$$

$$\therefore L(a \cos \theta, \pm \frac{b^2}{a}) \quad \downarrow$$

~~L is not correct~~

~~is not correct~~

Since $\sin \theta = \pm \frac{b}{a}$



$$\therefore \cos \theta = \frac{\sqrt{a^2 - b^2}}{a} \quad \downarrow$$

So $x = a \cos \theta$

$$= \pm a \frac{\sqrt{a^2 - b^2}}{a} = \pm \sqrt{a^2 - b^2} \quad \downarrow$$

for ellipse $b^2 = a^2(1 - e^2)$

$$\therefore a^2 - b^2 = a^2 e^2$$

$$\therefore x = \pm \sqrt{a^2 e^2} \\ = \pm ae \quad \downarrow$$

which is the foci

①) 5) i) $xy = 4$

$P(2p, \frac{2}{p})$

$y = \frac{4}{x}$

$y = -\frac{4}{x^2} = -\frac{1}{p^2}$ ✓

Normal: $\frac{p^2}{2}$

$y - \frac{2}{p} = p^2(x - 2p)$

$py - 2 = p^3x - 2p^4$

$py - p^3x = 2(1 - p^4)$ ✓

ii) $Q(2q, \frac{2}{q})$

$\frac{2p}{q} - 2p^3q = 2(1 - p^4)$

~~$\frac{2p}{q}$~~

$2p - 2p^3q^2 = 2q(1 - p^4)$ ✓

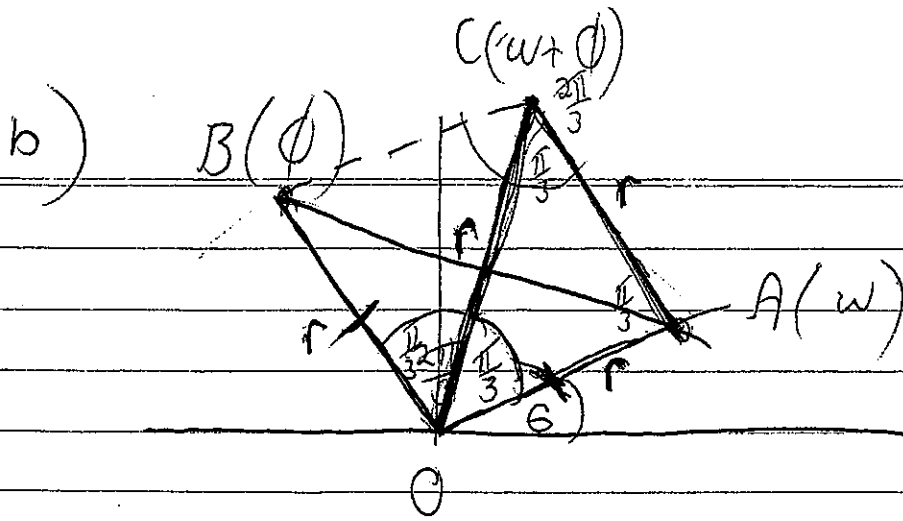
$p - p^3q^2 = q - qp^4$

$p - q = p^3q^2 - qp^4$

$-(q - p) = p^3q(q - p)$

$\therefore p^3q = -1$ ✓

4



$OACB$ is a rhombus $\therefore CO$ bisects $\angle BOA$

Let $\text{Arg } w = \theta$

$$\therefore \text{Arg } (w + \phi) = \left(\frac{\pi}{3} + \theta \right)$$

$$\text{Arg } (w + \phi)^2 = \left(2\frac{\pi}{3} + 2\theta \right)$$

$$|w + \phi| = r$$

$$\therefore \text{Arg } |(w + \phi)^2| = r^2$$

$$\therefore (w + \phi)^2 = r^2 \text{cis} \left(\frac{2\pi}{3} + 2\theta \right)$$

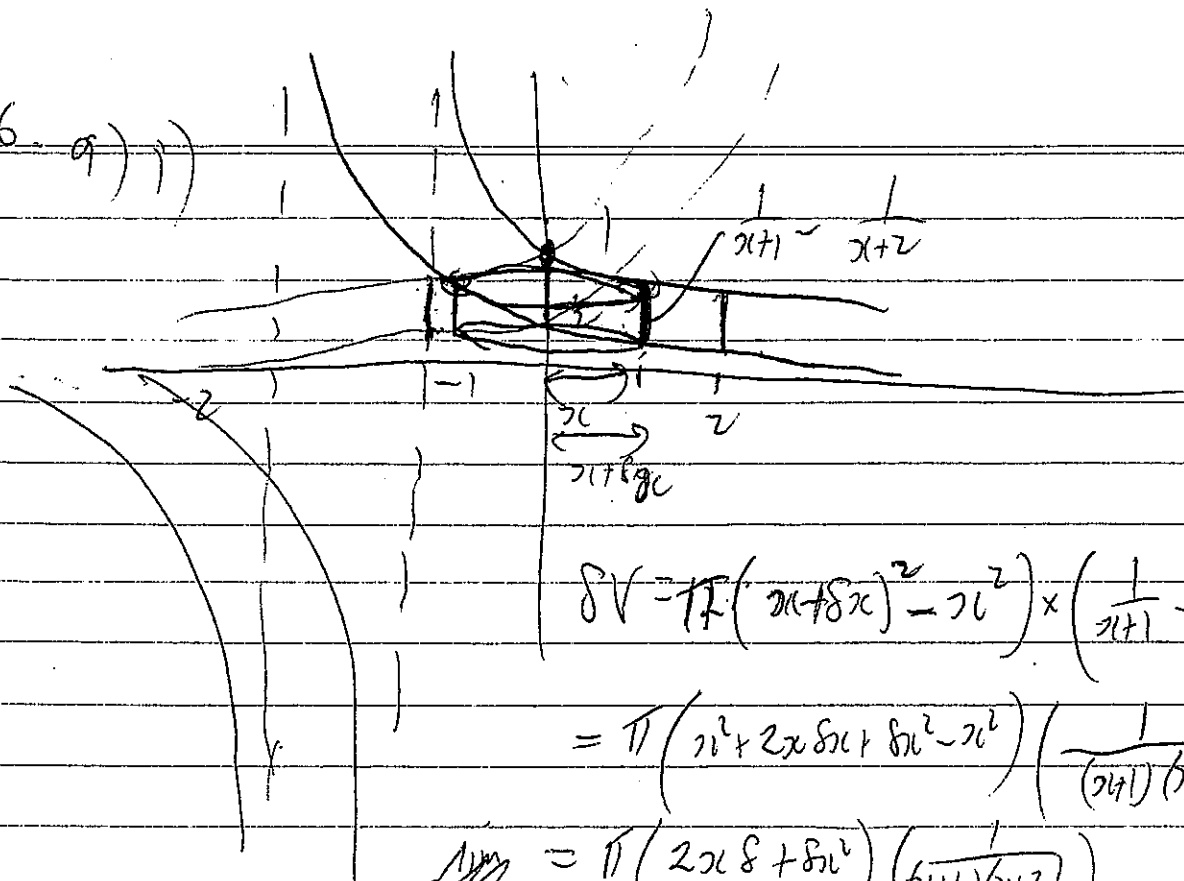
$$w\phi = r \text{cis } \theta \times r \text{cis} \left(\theta + \frac{\pi}{3} \right)$$

$$= r^2 \text{cis} \left(\theta + \theta + \frac{\pi}{3} \right)$$

$$= r^2 \text{cis} \left(2\theta + \frac{\pi}{3} \right)$$

$$= (w + \phi)^2$$

16. a) 1)



$$\delta V = \pi \left((x+\delta x)^2 - x^2 \right) \times \left(\frac{1}{x+1} - \frac{1}{x+2} \right)$$

$$= \pi \left(x^2 + 2x\delta x + \delta x^2 - x^2 \right) \left(\frac{1}{(x+1)(x+2)} \right)$$

$$\lim_{\delta x \rightarrow 0} \delta V = \pi (2x\delta x + \delta x^2) \left(\frac{1}{(x+1)(x+2)} \right)$$

$$\frac{dV}{dx} = \pi (2x\delta x + \delta x^2) \left(\frac{1}{(x+1)(x+2)} \right)$$

$\lim_{\delta x \rightarrow 0}$

$$\frac{dV}{dx} = \pi \int_0^2 2x \left(\frac{1}{(x+1)(x+2)} \right) dx$$

$$V = \pi \int_0^2 2x \times \frac{1}{(x+1)(x+2)} dx$$

$$= 2\pi \int_0^2 \frac{x}{(x+1)(x+2)} dx$$

$$\frac{a}{(x+1)} + \frac{b}{x+2} = \frac{a(x+2) + b(x+1)}{(x+1)(x+2)} = \frac{x}{(x+1)(x+2)}$$

$$2a + b = 0 \quad a + b = 1$$

$$\therefore a = -1 \quad b = 2$$

$$= 2\pi \int_0^2 \left(\frac{-1}{x+1} + \frac{2}{x+2} \right) dx = 2\pi \left(2\ln(x+2) - \ln(x+1) \right) \Big|_0^2$$

$$= 2\pi (2\ln 4 - \ln 3 - (2\ln 2 - \ln 1))$$

$$= 2\pi (\ln \frac{16}{3} - \ln 4)$$

$$= 2\pi \ln \left(\frac{4}{3} \right)$$

$$b) I_n = \int_0^{\frac{1}{2}} \frac{1}{(1+4x^2)^n} dx$$

$$I_1 = \int_0^{\frac{1}{2}} \frac{1}{1+4x^2} dx$$

$$= \frac{1}{4} \int_0^{\frac{1}{2}} \frac{1}{(\frac{1}{2})^2 + x^2} dx$$

$$= \frac{1}{2} \tan^{-1} 2x \Big|_0^{\frac{1}{2}}$$

$$= \frac{1}{2} (\tan^{-1}(1) - \tan^{-1}(0))$$

$$= \frac{1}{2} \times \frac{\pi}{4}$$

$$= \frac{\pi}{8}$$

$$1) \int_0^{\frac{1}{2}} \frac{1}{(1+4x^2)^n} dx = x \cdot (1+4x^2)^{-n} \Big|_0^{\frac{1}{2}} - (-n) \int_0^{\frac{1}{2}} \frac{8x}{(1+4x^2)^{n+1}} dx$$

$$= \frac{1}{2 \times 2^n} + 2n \int_0^{\frac{1}{2}} \frac{4x^0}{(1+4x^2)^{n+1}} dx$$

$$= \frac{1}{2^{n+1}} + 2n \int_0^{\frac{1}{2}} \frac{1+4x^2}{(1+4x^2)^{n+1}} - \frac{1}{(1+4x^2)^{n+1}} dx$$

$$= \frac{1}{2^{n+1}} + 2n \int_0^{\frac{1}{2}} \frac{1}{(1+4x^2)^n} - \frac{1}{(1+4x^2)^{n+1}} dx$$

$$I_n = \frac{1}{2^{n+1}} + 2n(I_n - I_{n+1})$$

$$I_n = \frac{1}{2^{n+1}} + 2nI_n - 2nI_{n+1}$$

$$\therefore \cancel{(1-2n)} (1-2n) I_n = \frac{1}{2^{n+1}} - 2n I_{n+1}$$

$$\therefore I_n = \frac{1}{2^{n+1}(1-2n)} - \frac{2n I_{n+1}}{1-2n}$$

$$= \frac{1}{2^{n+1}(1-2n)} + \frac{2n I_{n+1}}{2n-1}$$

$$\therefore 1) \quad I_1 = \frac{1}{2^2(1-2)} + \frac{2I_2}{1} = \frac{\pi}{8}$$

$$2I_2 = \frac{\pi}{8} + \frac{1}{4}$$

$$I_2 = \frac{\pi+2}{16}$$

$$I_2 = \frac{1}{2^3(1-4)} + \frac{4I_3}{3} = \frac{\pi+2}{16}$$

$$\frac{4I_3}{3} = \frac{\pi+2}{16} + \frac{1}{24}$$

$$\therefore I_3 = \frac{3}{4} \left(\frac{\pi+2}{16} + \frac{1}{24} \right)$$

$$= \frac{3}{4} \left(\frac{3\pi+6+2}{48} \right)$$

$$= \frac{9\pi+24}{4 \times 48} = \frac{3\pi+8}{64}$$